THE LONDON SCHOOL OF ECONOMICS AND POLITICAL SCIENCE

Spring 2024 Exam

MY457

Causal Inference for Observational and Experimental Studies

2023/24 Syllabus – not for resit/deferred candidates

<u>Instructions to candidates</u>

This paper contains five questions. Answer all questions.

Question 1 is worth 10 marks in total, Question 2 20 marks in total, Question 3 25 marks, Question 4 15 marks, and Question 5 30 marks. The marks in brackets indicate marks available for each part of each question.

You have been supplied with notepaper pages at the back of this question paper. **Notes made on this page will <u>NOT</u> be marked. Only notes made in an answer booklet will be sent for marking.**

Candidates must not take the question paper away with them after the examination. Please leave it on your desk at the end of the examination.

Time allowed	**Reading Time:**	*None*
	Writing Time:	*2 hours*
You are supplied with:		*No additional materials*
You may also use:		*No additional materials*
Calculators:		*Calculators are allowed in this exam*

1. (a) Explain the notation Y_{di} assuming that $D \in \{0, 1\}$ represents a binary treatment for a study in which we observe individuals $i = \{1, \ldots, N\}$. [2 marks]

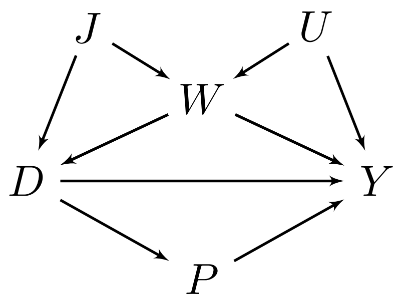
   (b) Given your answer to 1.a., explain the notation $Y_{0i}|D_i = 1$. [2 marks]

   (c) Contrast the meaning of $Y_{1i}|D_i = 0$ with the meaning of Y_{1i}. [2 marks]

   (d) When would $Y_{1i} = Y_{0i} = Y_i$, where Y_i is the observed outcome for unit i? [2 marks]

   (e) Contrast the meaning of $Y_{0i}|D_i = 1$ with the meaning of $Y_{0i}|D_i = 0$. [2 marks]

2. Consider the following Directed Acyclic Graph. For the next four questions, determine whether the proposed conditioning set is sufficient to identify the effect of D on Y.



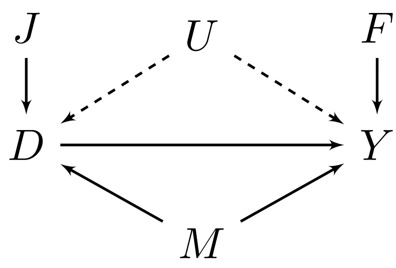
(a) Is the conditioning set $\{J\}$ sufficient? Why? [2 marks]

(b) Is the conditioning set $\{J, U, W, P\}$ sufficient? Why? [2 marks]

(c) Is the conditioning set $\{J, U\}$ sufficient? Why? [2 marks]

(d) Is the conditioning set $\{W, U\}$ sufficient? Why? [2 marks]

Now consider a new Directed Acyclic Graph, shown below. For the next two questions, assume we wish to identify the effect of D on Y.



(e) Explain the role U plays. Why are the edges U emits dashed? [3 marks]

(f) In your view, is M a good, bad, or neutral control? Why? [3 marks]

(g) In your view, is J a good, bad, or neutral control? Why? [3 marks]

(h) In your view, is F a good, bad, or neutral control? Why? [3 marks]

3. (a) The table below presents a hypothetical study involving twelve units. You can assume these units represent the entire population (i.e., set aside issues related to sampling uncertainty). Potential outcomes are given as Y_1 and Y_0 for treatment $D \in \{0, 1\}$, where 1 represents treatment and 0 control, and $X \in \{0, 1\}$ is a binary pre-treatment covariate.

Unit	Y_0	Y_1	D	X_1	X_2
3	22	27	1	0	0
4	21	23	1	0	1
5	7	5	0	0	0
8	21	22	0	0	1
9	16	19	0	0	0
11	26	32	0	0	0
1	30	25	1	1	1
2	18	20	1	1	0
6	14	15	1	1	0
7	27	26	0	1	1
10	18	22	0	1	1
12	15	18	0	1	0

i. Calculate the true average treatment effect (ATE), the true average treatment effect on the treated (ATT), and the true average treatment effect for the untreated (ATU). Be sure to show your working. [6 marks]

ii. Näively estimate the ATE using only observed outcomes and the treatment indicator D (do not condition on X_1 or X_2 in answer to this question). Be sure to show your working. [2 marks]

iii. Under what two assumptions is your answer in 3.a.ii an unbiased estimate of the true ATE? [2 marks]

(b) Consider again the data introduced in 3.a.

i. Calculate the propensity score for being assigned to the treatment group for each combination of X_1 and X_2. [4 marks]

ii. Using the values in the table, calculate the true conditional average treatment effects (CATEs) for each combination of X_1 and X_2. [4 marks]

iii. Explain the role X_1 and X_2 play in this study as relates to both the assignment mechanism and treatment effect moderation. [2 marks]

iv. Estimate the ATT using propensity score matching with replacement. If you have tied matches (many controls equivalently matched to one treated unit), you should take the average of all those control units. [5 marks]

4. (a) Using notation as appropriate, describe in your own words the basic idea, key elements, and identification assumption(s) of the **canonical two-period difference-in-differences design**. Explain what causal effects could potentially be estimated under this design, and how they might be estimated. In what kinds of settings would this design be most defensible? Does the design have any weaknesses? [8 marks]

   (b) Consider a difference-in-differences design in which we have **more than two time periods** and **treatment assignment** is **staggered** over time. Such settings are often analysed with a classical two-way fixed effects (TWFE) estimator. Explain what the TWFE estimator is and how it works. Discuss the key weakness of the TWFE estimator and comment on how **one** of the modern difference-in-differences estimators addresses these issues. [7 marks]

5. Imagine we wish to understand the effect of winning a university scholarship ($D \in \{0, 1\}$) on future earnings (Y). In our setting some students are randomly either eligible ($Z = 1$) or ineligible ($Z = 0$) for the scholarship. In the table below we are given the total number of students (units) observed for each combination of Z and D, as well as the average annual lifetime earnings for the eligible group and the ineligible group.

	Scholarship ($D = 1$)	No Scholarship ($D = 0$)	$\bar{Y}$
Eligible ($Z = 1$)	225	400	£56,000
Ineligible ($Z = 0$)	0	1250	£49,000

(a) Assuming SUTVA holds but with no other information about the study setting:

i. What research design might be appropriate in this setting? List the assumptions – aside from SUTVA – on which the design rests. [5 marks]

ii. Estimate the intention-to-treat (ITT) effect using the data provided in the table. [2 marks]

iii. Estimate the local average treatment effect (LATE) using only the data provided in the table. [4 marks]

iv. Contrast your estimate of the ITT against your estimate of the LATE. Are they different, or the same? Why? In your answer, be sure to carefully explain what each estimand represents. Finally, explain some of the advantages and limitations of these two estimands compared to the ATE. [4 marks]

(b) After further investigation it turns out that eligibility for the scholarship is not in fact random, and is more complicated than first assumed. When an individual scores over 80% on a pre-entry assessment test, they become eligible for the scholarship. If they score 80% or less they remain ineligible.

i. Given this new information, would the original design be appropriate? Explain your answer. [2 marks]

ii. What research design would you propose instead? In what ways does this new research design improve upon the old one? [3 marks]

iii. Explain at least one weakness of the new design. [2 marks]

iv. A key identifying assumption in this design is often referred to as 'continuity in potential outcomes.' Explain in words what this means, and describe one test you might conduct to falsify the identifying assumption. [2 marks]

(c) Draw two graphs that demonstrate the intuition behind the new design discussed in (5.b). The first graph should have the pre-entry test score on the x-axis, and average take-up rate of the treatment (the conditional mean of D) on the y-axis. The second graph should have the pre-entry test score on the x-axis, and average annual lifetime income (the conditional mean of Y) on the y-axis. You do not need to use real data (you can just make up points/lines), but do try to show any jumps that might occur in the y-axis variables somewhat accurately, given the data in the table above. [6 marks]

END

Notes made on this page will NOT be marked.
Only notes made in an answer booklet will be sent for marking.

Notes made on this page will NOT be marked.
Only notes made in an answer booklet will be sent for marking.

Notes made on this page will <u>NOT</u> be marked.
Only notes made in an answer booklet will be sent for marking.

Notes made on this page will <u>NOT</u> be marked.
Only notes made in an answer booklet will be sent for marking.